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Primer

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Acute rheumatic fever and rheumatic heart disease

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Abstract

Acute rheumatic fever (ARF), the acute autoimmune sequela of group A *Streptococcus* infection, and its chronic life-threatening manifestation, rheumatic heart disease (RHD), which causes permanent heart valve damage, remain major drivers of preventable cardiovascular mortality and disability in low-income and middle-income countries. Since the mid-1990s, multidisciplinary and international collaborative research has yielded new insights into the role of superficial skin infections such as impetigo in ARF pathogenesis and the disease mechanisms that underlie the progression from ARF to RHD, and has explored novel treatments and delivery mechanisms in clinical trials and progressed the development of Strep A vaccines. Early detection and patient care have advanced, with revised echocardiographic screening criteria and the 2024 WHO guidelines, which provide a framework for evidence-based management and therapeutic strategies. Notwithstanding these advancements, considerable disparity persists in the care, control and treatment of advanced disease, necessitating coordinated programmes, prioritization by global agencies and local governments, and substantial involvement of those with RHD. Of note, few to no cardiac surgery programmes exist in some countries most burdened by RHD. Future research must encompass implementation studies for validated evidence-based therapies, translational technologies and applications, together with a definitive pathway to direct patient effect, to achieve the WHO target of eliminating RHD as a public health problem.

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Introduction

Acute rheumatic fever (ARF) is a rare autoimmune complication that can develop following group A *Streptococcus* infection (now referred to as Strep A). Rheumatic heart disease (RHD) is a life-threatening sequel that results from chronic heart valve damage following single or multiple recurrent, but often undocumented, under-recognized episodes of ARF.

ARF occurs 7–21 days after a Strep A infection of the throat or skin. Although Strep A throat infections have traditionally been considered the primary antecedent, emerging evidence has provided important insights into the role of skin infections (such as impetigo – a highly contagious skin infection that causes skin sores or blisters around the nose, mouth, hands and feet) in the development of ARF in certain geographical regions[1]. Although Doppler echocardiography was incorporated into the 2015 revised Jones criteria, the diagnosis of ARF using the revised criteria remains an imperfect clinical decision tool[2]. Clinical features such as chorea, carditis, arthritis, subcutaneous nodules and the classical erythema marginatum skin rash need to be supplemented with evidence of prior Strep A infection to establish a diagnosis, which also requires microbiological and serological laboratory capacity that is often not available in low-resource settings (Table 1). ARF often manifests with subtle symptoms that can be easily misdiagnosed and it is more prevalent in Africa than previously recognized, underscoring the urgent need for improved diagnostic strategies, including molecular testing[3].

RHD remains one of the major causes of acquired heart disease in children, adolescents and young people living in low- and middle-income countries (LMICs), marginalized communities and low-resource settings in high-income countries (HICs)[4]. In addition, most cases in RHD-endemic countries present late[5] with complications such as cardiac failure, atrial fibrillation, thrombosis, stroke and infective endocarditis, all of which cause substantial morbidity and mortality[6].

Since the early 2000s, substantial progress has been made in our understanding of the pathophysiology of RHD. The World Heart Federation (WHF) guidelines for echocardiographic diagnosis of RHD were revised[7], and the first randomized trial in Ugandan schoolchildren to investigate the role of secondary prophylaxis (that is, long-term penicillin to prevent ARF recurrences and disease progression or worsening) was conducted[8]. A resurgence of research in ARF and RHD has been rewarded by new funding into immunological correlates of protection[9], development of a diagnostic test for ARF, use of artificial intelligence for RHD echocardiography screening[10], vaccines for Strep A and implementation science with local communities[11].

Regional offices of global agencies such as the WHO, have empowered community-led education and advocacy in partnership with initiatives such as Reach (Rheumatic Heart Disease, Evidence Advocacy Communication and Hope; a non-governmental organization focused on providing in-country scientific and technical support), which provides technical support to countries through decision-support tools, algorithms and low-literacy advocacy resources for implementation of RHD control programmes, as well as the WHF's 'Colours to Save Hearts' campaign. The first ever global RHD conference in 2023 brought together researchers, advocates, patients and communities to share research and their lived experience and consider new avenues to progress the advances in this field. However, despite all the progress[4], the stark reality of high mortality in high-risk populations remains[12], with continued lack of medical, surgical and interventional care for those with severe disease. In addition, geopolitical instability reminds us anew of the fragility of health systems in LMICs with persisting inequalities and the need to focus activities on the most vulnerable[13,14]. We need

Table 1 | The 2015 revised Jones criteria for diagnosis of ARF

Category	Population		Notes
	Moderate and high risk	Low risk	
Streptococcal evidence	Evidence of preceding Strep A infection	Evidence of preceding Strep A infection	Highest evidence is a rising streptococcal antibody titre, followed by a single elevated titre or a positive throat culture, rapid antigen test or molecular POC test in a child with history of sore throat
Major criteria	Carditis (clinical or on Doppler echocardiography)	Carditis (clinical or on Doppler echocardiography)	Occurs in >50% of cases of acute ARF; mitral regurgitation occurs more commonly than aortic regurgitation[96]; rarely, myocarditis and/or pericarditis and not in isolation; early disease commonly detectable via echocardiography; diagnosed by the 2023 WHF criteria for RHD (in the absence of better guidance)[7]
	Polyarthritis only	Polyarthritis, polyarthralgia, monoarthritis	Affects 60–80% of cases of acute ARF; most common presentation is migratory, asymmetrical polyarthritis of large joints; polyarthralgia and monoarthritis more common in high-risk areas
	Chorea (Sydenham's)	Chorea (Sydenham's)	Involuntary movements, neuropsychiatric signs; can occur alone without other criteria[1]
	Erythema marginatum	Erythema marginatum	Rare; pink, spreading macular rash, brought out in warm temperatures (such as a bath), might be hard to see in pigmented skin
	Subcutaneous nodules	Subcutaneous nodules	Rare; small, painless lumps, over extensor surfaces, often with carditis
Minor criteria	Fever (≥38.5°C)	Fever (≥38.0°C)	Occurs in >90% of cases of acute ARF
	Polyarthralgia	Monoarthralgia	Only counted if joints are not a major manifestation
	ESR ≥60 mm/h and/or CRP ≥30 mg/l	ESR ≥30 mm/h and/or CRP ≥30 mg/l	Often elevated but nonspecific
	Prolonged PR interval on ECG (accounting for age)	Prolonged PR interval on ECG (accounting for age)	Only counted if carditis is not a major manifestation

For more information, see ref. 2. ARF, acute rheumatic fever; CRP, C-reactive protein; ECG, electrocardiography; ESR, erythrocyte sedimentation rate; POC, point of care; RHD, rheumatic heart disease; Strep A, group A *Streptococcus*; WHF, World Heart Federation.

coordinated action by governments, donors and international agencies with urgent investment in universal access to antibiotics, integration of RHD screening into primary and antenatal care, and prioritization of vaccine development.

In this Primer, we summarize key developments in the epidemiology, pathophysiology, diagnostic approaches, screening and prevention and clinical management of individuals with ARF and RHD. We conclude with an overview of quality of life, from the perspective of a frontline health worker and lived experience expert and propose future areas of impactful research.

Epidemiology

Incidence and prevalence

The incidence of ARF is highest in children 5–14 years of age[15]. A systematic review from 2008 found a mean incidence of 19 per 100,000 (range 5–51), albeit with no data found from sub-Saharan African countries[16]. Another study using the 2015 Jones criteria estimated the incidence of definite ARF in children 5–14 years of age at 13–25 per 100,000 in two districts in Uganda[3]. Given the documented RHD prevalence of 1.5% in Uganda[17], this incidence is much lower than expected, considering that every case of RHD is preceded by a case of ARF. These data suggest that ARF is underdiagnosed in many endemic regions. Studies have found higher incidence in countries with more developed health systems and resources for earlier diagnoses and tracking cases, with age-specific incidence of 108 per 100,000 or more (in children 5–14 years of age) documented in indigenous populations of Australia[15,18]. ARF is often underdiagnosed and inadequately treated owing to scarce diagnostic tools, limited resources to initiate and sustain secondary prophylaxis and barriers to health-seeking behaviour in high-burden settings[3,15]. Additionally, many cases of RHD develop in the absence of overt ARF because of low-grade carditis, which might explain some of the discrepancy between the low incidence of ARF and the high prevalence of RHD in the same community.

The Global Burden of Disease (GBD) study estimated that 55 million people were living with RHD worldwide in 2021, but prevalence ratios varied greatly across countries and regions[19] (Fig. 1). Overall, in countries where RHD is endemic, its prevalence consistently ranges between 0.5% and 2% in children and young adults[20–22]. The highest prevalence of RHD is estimated to be in sub-Saharan Africa (age-standardized ratio of 1,370 per 100,000), followed by South Asia (724 per 100,000) and then the Middle East and North Africa (603 per 100,000). However, these national and regional estimates hide the high burden of ARF and RHD in some sub-populations, including Indigenous people living in HICs such as Australia and New Zealand[19]. Most people with RHD are from populous countries in South Asia and sub-Saharan Africa[23] (Fig. 2). The total number of individuals living with RHD in endemic countries has steadily increased since the mid-1990s[24] owing partially to persistently high fertility rates and decreasing child mortality from other causes of death. Without large and rapid reductions in Strep A infection and ARF, demographic trends will continue to keep RHD prevalence counts high in the coming decades[19]. Accuracy of worldwide estimates for RHD prevalence has greatly improved since the widespread implementation of screening echocardiography, as demonstrated by current GBD results, which incorporated 55 prevalence studies in endemic settings[25].

Mortality and morbidity

In 2021, the GBD study estimated 373,000 deaths from RHD, corresponding to a mortality rate of 4.5 per 100,000 across all age groups[19]. After adjusting for age, South Asia was estimated to have the highest RHD mortality (14.6 per 100,000), but mortality was estimated to be below the average global rate in sub-Saharan African regions[19] (Fig. 1). Further, these estimates tend to underestimate deaths in children and young adults and overestimate deaths in older adults[26]. One of the main reasons these mortality estimates are low relative to the high disease prevalence in this region might be the lack of population cause-of-death data from vital registration systems[27]. Verbal autopsy, the method used to ascertain cause of death in these countries, is not specific for RHD and probably leads to misclassification of RHD-related deaths to other cardiovascular causes[24]. Contributing to this error is the lack of specificity of the International Classification of Diseases, 10th edition (ICD-10) codes for the detection of RHD documented in high-income settings where the disease is substantially less frequent[28]. Record-linkage studies in Fiji found significantly higher age-specific mortality than was estimated in the GBD studies[29]. High-quality studies of RHD mortality in the general population in RHD-endemic countries is crucial to enable accurate estimates of RHD mortality by GBD analyses.

Hospital-based data, albeit not representative of the population, have shown high morbidity associated with RHD[12]. Complications peak in the third decade and result in high early mortality, at 116 per 1,000 patient-years and a median age of 28 years in endemic countries. However, it is important to note that these estimates come from studies of people presenting with severe forms of RHD[30]. Population-based studies using record linkage have shown that heart failure is the most commonly reported complication — 16.5% were admitted with heart failure over a 5-year follow-up (translating to 2.5 per 100,000 person-years), followed by atrial fibrillation (12.7%) and a smaller proportion of patients for stroke (4.9%) and infective endocarditis (3.4%)[31]. Admission for any complication is associated with a fivefold increased risk of death (HR 5.4, 95% CI 3.4–8.8; $P < 0.001$)[31].

RHD risk factors

Although the causal relationships between Strep A infection and the development of ARF and RHD are well established[15], the risk factors have yet to be fully understood. Many studies have documented associations between Strep A infection, ARF and RHD and one or more host and environmental factors[32] (Fig. 3), but how these factors work together is less clear and the quality of the evidence is low.

Host factors. Host susceptibility leading to RHD encompasses demographic and genetic factors. The finding that the incidence of ARF is highest in children 5–14 years of age is indicative of the part that social interactions play as a risk factor during this developmental stage[33]. Studies have documented a female predominance for RHD in LMICs[21] and in high-prevalence communities in HICs[18]. This observation is postulated to be linked to autoimmune susceptibility and increased health-seeking behaviour. Genetic factors have an important role in ARF susceptibility — only 3–6% of individuals are predisposed to development of ARF after a Strep A infection[34], but the inheritance pattern is complex and not fully understood. Candidate gene studies spanning decades have yielded heterogeneous results implicating HLA locus and others, but have not been replicated across settings[35]. Findings from genome-wide association studies (GWAS) are varied; two studies identified HLA regions as major risk factors and a third study identified *IGHV4-61* as an important risk factor[36–38]. One study estimated that RHD has a polygenic heritability of 0.49 in African individuals with a risk locus on chromosome 11 that reached genome-wide significance[39]. This study also replicated the *IGHV4-61*02* allele from Oceania countries in a meta-analysis of Black and admixed African individuals, supporting its

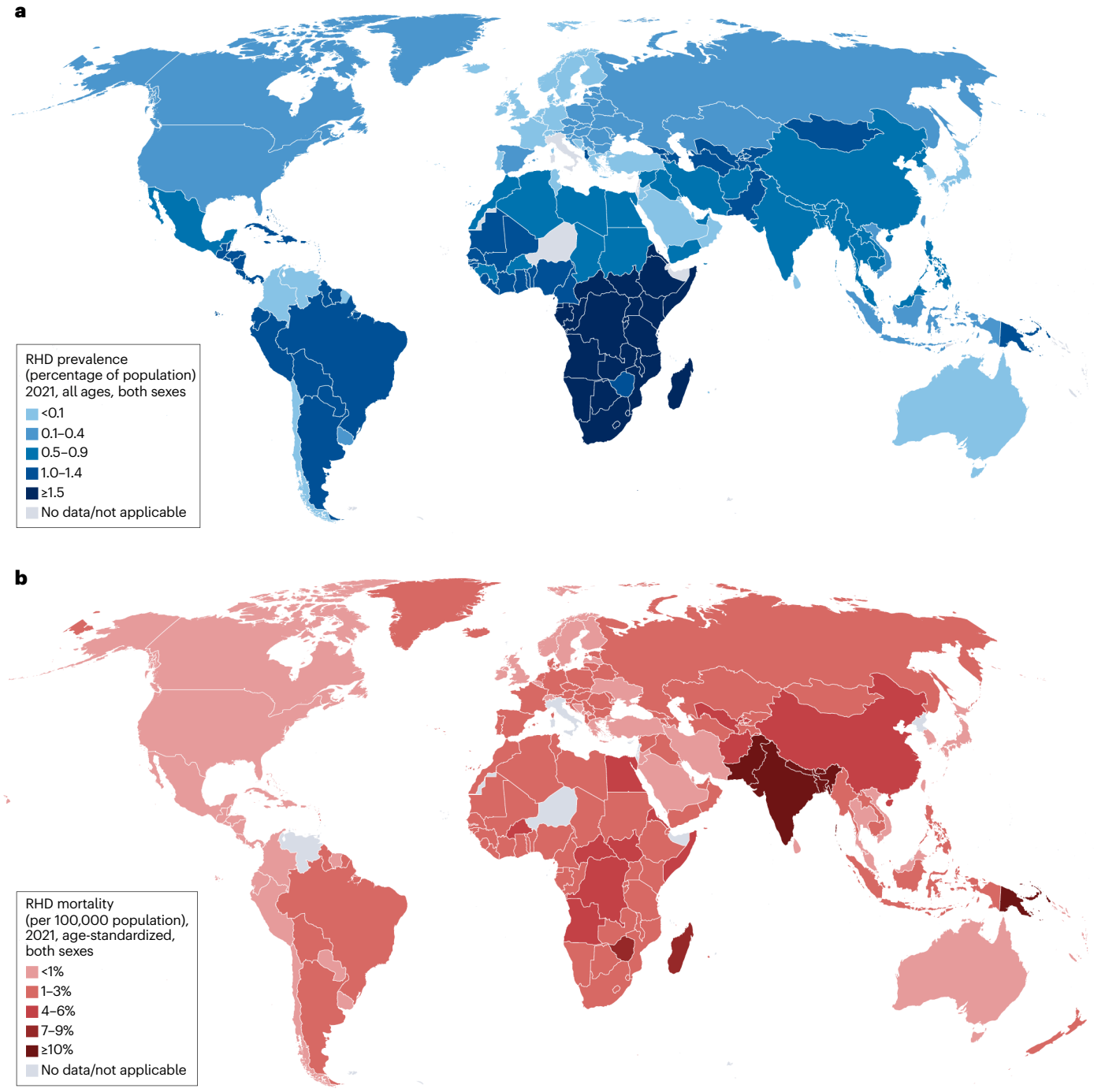


Fig. 1 | RHD prevalence and mortality. a, Worldwide prevalence of rheumatic heart disease (RHD). **b**, Mortality due to RHD. In 2021, there were an estimated 373,000 deaths from RHD, corresponding to a mortality rate of 4.5 per 100,000 across all age groups. Adjusted for age, South Asia was estimated to have the highest RHD mortality (14.6 per 100,000), whereas mortality was estimated to be below the average global rate in sub-Saharan Africa.

role in susceptibility to RHD[37,39]. Further work, including meta-analyses, is needed to better understand the genetic component of disease susceptibility in diverse endemic populations.

Environmental factors. People living in settings that promote easy spread of pathogens are more likely to acquire Strep A infection, a finding supported by studies that show that Strep A strain types are highly shared between close contacts in studies[40]. Thus, high social interactions and large households plausibly influence disease risk in children 5–14 years of age[41]. Household crowding is among the most commonly reported environmental risk factor (OR 3.88, 95% CI 1.68–8.98), has been found to be directly correlated with Strep A carriage[42]

and might be the factor that links poverty to RHD prevalence[43]. Studies have described several other environmental and social factors in association with RHD antecedents and include housing conditions, health status including nutrition, socioeconomic factors such as low education level, low income and limited access to healthcare for the treatment of Strep A[33,43]. Of note, superficial skin infections such as impetigo are postulated to be the link between skin infections and ARF and have been demonstrated to confer an increased risk for ARF and/or RHD in Pacific and Indigenous Australian communities[44]. In addition, other unusual factors have been described, including consumption of sugar-sweetened beverages (OR 2.0, 95% CI 1.13–3.54)[32]. However, many studies were observational, of low to medium quality, which have yielded heterogeneous findings. Thus, many studied factors generally had mixed or inconsistent results, with limited generalizability[43].

Mechanisms/pathophysiology

Pathogenesis of ARF and RHD

ARF is an autoimmune response that follows a local Strep A infection. Epidemiological and serological evidence supports the requirement for repeated Strep A infections to 'prime' the immune system for loss of tolerance and immune dysregulation in ARF[1,45]. Although outbreaks of ARF following documented Strep A pharyngitis have long established the role of pharyngitis as a preceding infection, skin infections are also increasingly recognized to play a part, either directly or as priming infections early in life, particularly given the high incidence of Strep A skin infections in many high-burden ARF settings[1,46,47].

Following a superficial Strep A infection, neutrophils, macrophages and dendritic cells phagocytose bacteria and present antigen to T cells. Both B cells and T cells respond to the Strep A infection via antibody production (IgM, IgA and IgG) and through T cell activation (mainly $CD4^+$ cells). In susceptible individuals, the host response to Strep A will result in breakdown of tolerance and an autoimmune reaction against host tissues, for example, the heart, brain, joints and/or skin[4] (Fig. 4). Molecular mimicry between Strep A antigens and similarly structured human proteins has been hypothesized to trigger these auto-reactive responses[48], with studies implicating the Strep A M protein and group A carbohydrate in this process. Evidence of molecular mimicry comes from cross-reactive monoclonal antibodies and T cells isolated from patients with ARF[49,50], as well as a Lewis rat model in which vaccination with adjuvanted heat-killed Strep A or Strep A antigens can result in valvulitis[51]. However, the pathophysiological link between Strep A infections and the development of ARF is the subject of ongoing investigation. The role of group A carbohydrate in ARF pathogenesis has been questioned owing to the quality of antigens used in older studies[52], and the most frequent phenotype in the rat model is myocarditis rather than valvulitis[53], raising questions as to model validity. Furthermore, cardiac myosins are proposed as a major target of cross-reactive antibodies, but as intracellular antigens predominantly localized to myocardium, are unlikely to be sufficient to initiate disease pathology. Studies have reported several other autoantigens in the pathogenesis including collagens, tropomyosins and *N*-acetylglucosamine (GlcNAc) glycosylation on host antigens[4,48,54], but a definitive autoantibody trigger remains elusive.

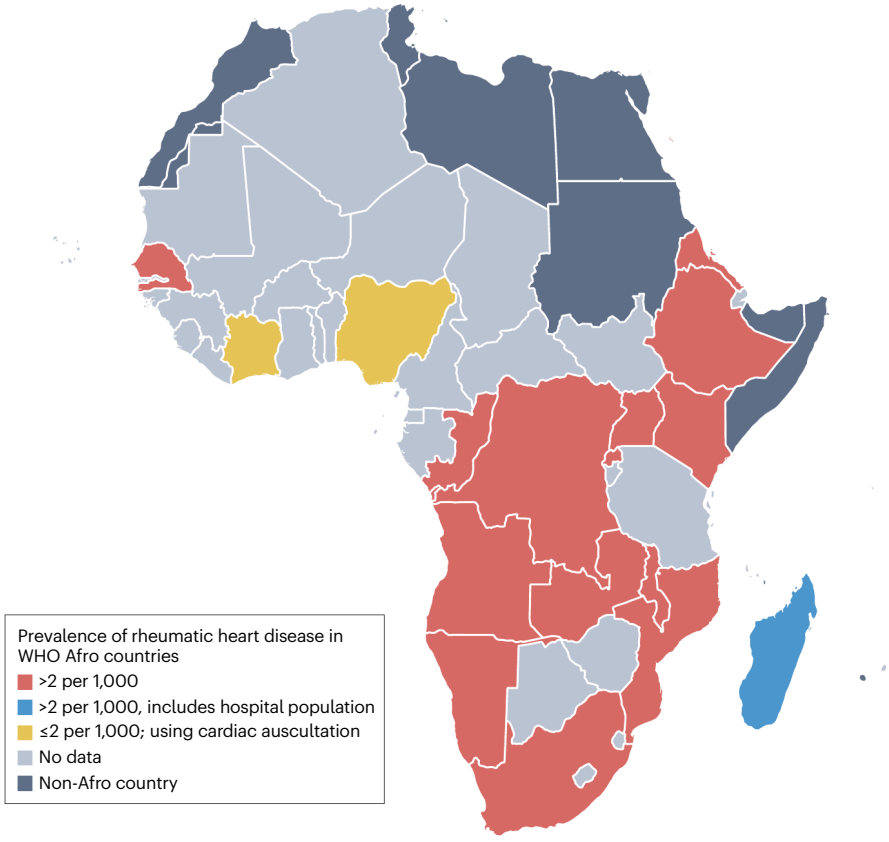


Fig. 2 | Prevalence of RHD in WHO-Afro countries. The prevalence of rheumatic heart disease in WHO-Afro region countries according to a review of systematic and literature reviews. Red depicts countries with a prevalence of >2 per 1,000 population, yellow shows regions with prevalence of <2 per 1,000 population. Blue includes community screening and yellow still includes cardiac auscultation. There were no data on prevalence in the countries shaded light grey, and countries in dark grey are not in the WHO-Afro region. The full publication with methodology can be found on the WHO-Afro website[23]. RHD, rheumatic heart disease.

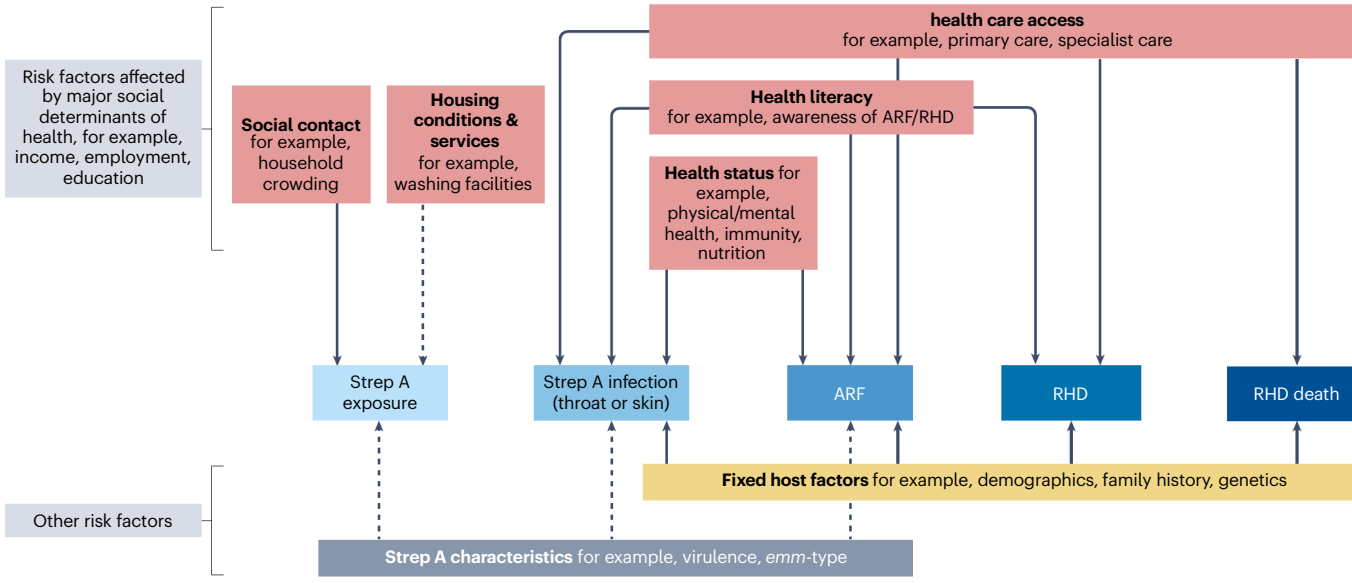


Fig. 3 | Hypothesized pathways by which social determinants of health influence the risk of exposure to Strep A and progression to ARF and RHD. There are multiple intersecting pathways by which social determinants of health are hypothesized to influence the risk of exposure to group A *Streptococcus* (Strep A) and further progression to acute rheumatic fever (ARF) and rheumatic heart disease (RHD). The major social determinants, together with pathogen and host factors act on the pathogenic pathway through social contact, housing conditions, health status and literacy and access to care. Adapted with permission from ref. 113, BMJ.

Following the triggering of immune dysregulation, pro-inflammatory immune responses drive disease activity in ARF. This process includes T helper (T_H) cells (T_H1 and T_H17 cells, dominated T cell responses[55]), inflammatory cytokines such as IL-6 and tumour necrosis factor (TNF)[56,57] and potent IgG3 polarized antibody responses[56,58,59], which contribute to immune-mediated tissue damage in affected tissues.

Pathogen factors

The Strep A strain type influences host interactions and immune responses that result in autoimmunity and the development of ARF. Molecular epidemiological studies have typically relied on single-locus *emm* typing, which analyses variability in the N-terminal region of the *emm* gene encoding the M protein[60]. Historically, a limited number of strains (*emm* types) were associated with ARF and were known as 'rheumatogenic' strains[61]. However, these strains are much less prevalent in LMICs and settings in which ARF persists today. In ARF-prevalent regions, very high strain diversity is observed, with a large number of different *emm* types co-circulating[62–67], raising doubts as to the relevance of the rheumatogenic strain concept for contemporary ARF[68]. The change in strain association suggests that some of the characteristics of Strep A reported in ARF outbreaks in the USA in the 1950s may be less relevant for current pathogenesis models. Indeed, the previously identified B cell and T cell epitopes implicated in molecular mimicry were not observed in >75% of ARF-associated strains in a global meta-analysis[68], pointing to the need for further research to identify components of Strep A that trigger ARF today.

Immune response and cytokine imbalance

An integral role for the immune system in disease development is suggested by the genetic association of RHD (via GWAS) with system components of the immune system including *IGHV4-61* (encoding immunoglobulin heavy chain)[37] and HLA regions[36,38]. Given that immunoglobulin genes underpin antibody production by B cells and HLAs present antigen to T cells, this points to involvement of B cells and T cell arms of adaptive immunity in the development of ARF and progression to RHD. Indeed, one study observed antibody and complement deposits on excised rheumatic valve tissue[69], confirming that antibodies contribute to ARF pathology. Serological studies have shown that children with ARF have experienced multiple Strep A infections before diagnosis with first-episode ARF[45]. This 'immune priming' process results in an anti-Strep A antibody repertoire in ARF that is increased in both magnitude and breadth compared with children with precursor skin and throat infections[70]. Striking increases in total IgG3 have also been observed in ARF, with antibody responses to the Strep A M protein being highly polarized to IgG3 (refs. 56,58). As IgG3 is the most potent immunoglobulin subclass with respect to downstream effector functions[59,71], repeated Strep A infections may amplify IgG3 dysregulation, which contributes to a loss of tolerance in ARF.

Analysis of human antigen serum reactivities in ARF has tended to use targeted approaches focused on coiled-coil antigens including members of the myosin and collagen protein families. Autoantibody binding within the human cardiac myosin heavy chain has been mapped to the S2 sub-fragment region within the myosin protein[72]. As cardiac myosin is an intracellular antigen, cardiomyocyte lysis would be required to enable antigen exposure to antibodies. Therefore, additional extracellular antigens must contribute to both initiation and perpetuation of the autoimmune response. Collagens are highly abundant proteins in the body and important structural proteins in heart valves. Studies have observed autoantibodies to collagens in patients with ARF[73–75], and one study has demonstrated that Strep A can aggregate collagen during infection, which could expose otherwise cryptic

epitopes and contribute to autoimmunity[76,77]. Collagen epitopes may also be exposed by inflammation-mediated tissue damage during carditis; studies have reported an abundance of COL1A1 and COL1A2 in RHD valves, and collagen epitopes have been shown to stimulate $CD8^+$ T cells from patients with RHD[78]. The application of high-content-array technologies to ARF, in which thousands of autoantigens are probed for serum reactivity in parallel, showed marked heterogeneity in autoantigen targets between ARF cases[74,79]. This finding suggests the involvement of epitope spreading, whereby an existing immune response is broadened by the exposure of new antigen targets resulting from earlier tissue damage, driving further inflammatory disease[74,80].

Cytokines are pivotal in orchestrating immune responses. In the context of ARF and RHD, imbalance between pro-inflammatory and anti-inflammatory cytokines contributes to disease progression. Key pro-inflammatory cytokines implicated include IL-6, TNF and IL-1β[56,57,81], which facilitate the recruitment and activation of immune cells and perpetuate inflammation[81] (Fig. 5). Histological analyses of RHD-affected heart valves reveal increased expression of T_H1 and T_H17 cell-associated cytokines, including IFNγ and IL-17, indicating a skewed T_H cell response[82]. T_H1 cells produce IFNγ, which activates macrophages and promotes inflammation, whereas T_H17 cells secrete IL-17, contributing to neutrophil recruitment and further inflammation. In keeping, persistent production of IL-1β in ARF has been shown to lead to the expansion of granulocyte–macrophage colony-stimulating factor (GM-CSF)-producing T_H1 cells, which in turn will amplify the inflammatory response[83]. Furthermore, regulatory T (T_{reg}) cells, which serve to control inflammation and self-reactive responses, are reduced in ARF and RHD[84]. Thus, expansion of self-reactive T_H1- or T_H17-skewed cells, combined with reduced T_{reg} cells, contribute to a sustained loss of tolerance, inflammation and immune-mediated tissue injury in ARF and RHD.

Chemokines, fibrosis and histopathological features

Immune-mediated tissue damage is contingent on honing of immune molecules and immune cells to these sites. Chemokines are integral to this process and stimulate the migration of immune cells via cognate cell receptors. The chemokines C-X-C motif chemokine ligand 9 (CXCL9) and CXCL10 are elevated in ARF and RHD and have roles in T cell recruitment to the heart[85]. Notably T cells expressing the CXCR3 receptor (binds to the chemokines CXCL9, CXCL10 and CXCL11) are reduced in circulation in ARF[84], and have been observed in RHD tissue[85], providing strong evidence for T cell honing and contribution to immune-mediated tissue damage. The adhesion molecule VCAM1 is thought to facilitate this T cell infiltration into heart valves in RHD, with its expression identified in rheumatic valvular endothelium and not in normal heart valve controls[86]. Additionally, TGFβ, a cytokine associated with fibrosis, is upregulated, contributing to the fibrotic remodelling of heart valves[55,81] (Fig. 5).

Aschoff bodies are nodules found in RHD heart tissue that consist of fibrinoid necrosis (death of small blood vessels) surrounded by T lymphocytes, plasma cells and activated macrophages known as Anitschkow cells[87]. Aschoff bodies are considered characteristic of ARF–RHD and are typically found in the myocardium[88]. Auto-reactive T cells are thought to perpetuate inflammation within the heart, contributing to fibrotic scarring and the progressive destruction of valve tissue[72]. These fibrotic processes seem central to cardiac damage

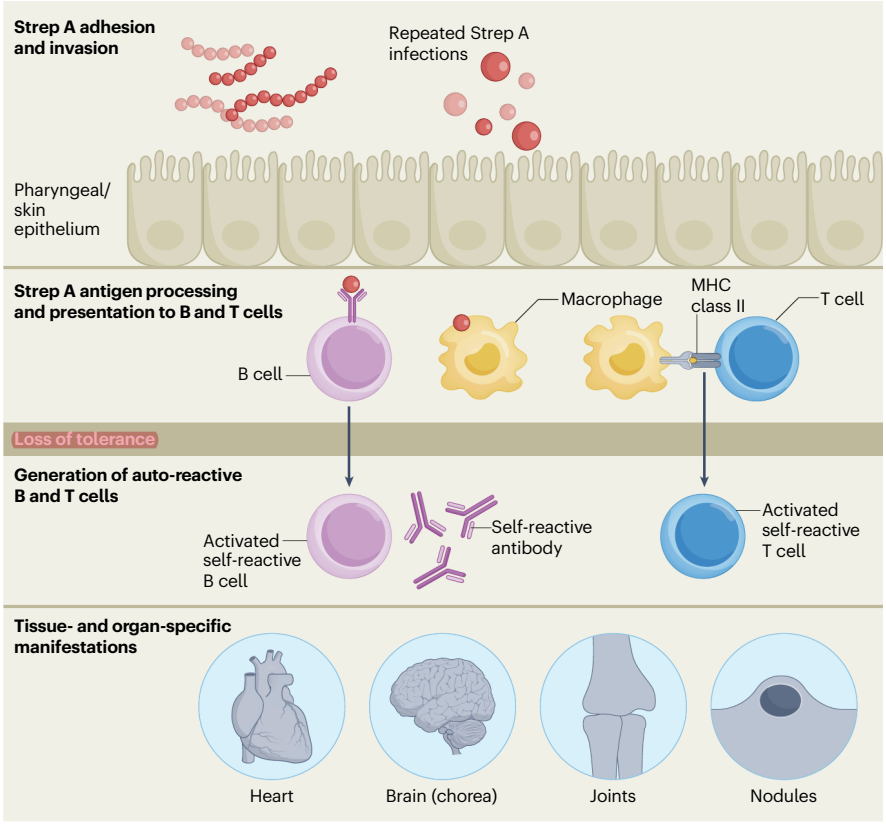


Fig. 4 | Generation of an auto-reactive immune response. Following repeated infections with group A *Streptococcus* (Strep A), adhesion and invasion of pharyngeal epithelial cells activates both B and T cells. A loss of tolerance, together with molecular mimicry, likely contribute to an autoimmune response, which results in the major manifestations of acute rheumatic fever (ARF). MHC, major histocompatibility complex. Adapted from ref. 4, Springer Nature Limited.

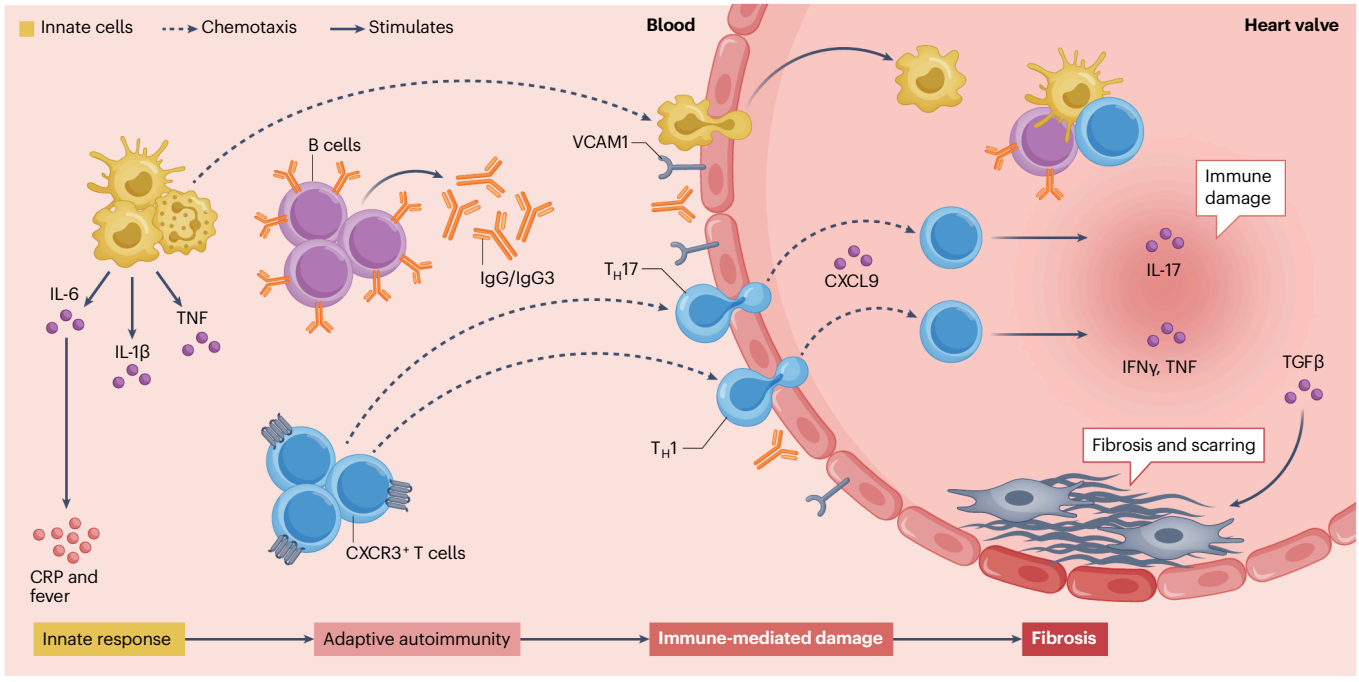


Fig. 5 | Cytokine- and chemokine-mediated autoimmunity in blood and heart valve tissue during ARF and RHD. Immune-mediated damage in acute rheumatic fever (ARF) and rheumatic heart disease (RHD). During the innate response, circulating cytokines IL-1β and tumour necrosis factor (TNF) increase pro-inflammatory cytokine secretion, immune cell migration and adhesion-molecule expression, including that of VCAM1. Circulating IL-6 stimulates production of C-reactive protein (CRP) and induces fever, and chemokine-induced chemotaxis results in neutrophil and/or monocyte infiltration of valve tissue. Circulating IgG3 is increased and the adaptive immune response produces heart antigen-specific B cells and T cells, with CXCR3-mediated chemotaxis, modulated by chemokines such as C-X-C motif chemokine ligand 9 (CXCL9), driving immune cell infiltration of heart valve tissue. In the valve, differentiated T helper 17 (T_H17) cells produce IL-17 and T_H1 cells produce IFNγ and TNF, which together cause immune-mediated structural damage. Subsequent tissue repair results in TGFβ-induced fibrosis and scarring. Cycles of inflammatory tissue damage and fibrotic scarring progressively damage heart valves, resulting in progression to RHD. Adapted with permission from ref. 81, Elsevier.

in RHD. Tenascin C, which is expressed following tissue injury and remodelling, is observed in RHD tissue in close proximity to inflammatory infiltrate, pointing to a mechanistic link between inflammatory processes and fibrosis[89]. This vicious cycle of damage is especially pronounced in the left-sided heart valves, which endure the highest pressure gradients and have a limited capacity for repair under persistent mechanical stress[90].

Taken together, repeated Strep A infections result in abnormal antibody responses, elevated total IgG3 and autoantibodies to cardiac antigens generated via a loss of tolerance. The disease is further sustained by a pro-inflammatory cytokine environment, T_H1 and/or T_H17 cell expansion and reduced numbers of T_{reg} cells, leading to chronic inflammation, epitope spreading and immune-mediated tissue damage. This damage is further compounded by defective cycles of fibrotic repair, resulting in heart valve scarring (Fig. 5).

Diagnosis, screening and prevention

Diagnosis

Acute rheumatic fever. Currently, no definitive test is available to diagnose ARF. Although several global research networks are actively working to identify reliable biomarkers: Searching for a Technology-Driven Acute Rheumatic Fever Test (START)[91], Acute Rheumatic Fever Diagnosis Collaborative Network (ARC) and the Preventing Rheumatic Injury biomarker Alliance (PRIMA), diagnosis currently relies on clinical criteria — most notably the modified Jones criteria (Table 1) — which first require exclusion of other conditions[2]. The 2015 revision of the Jones criteria has incorporated echocardiography to detect clinical and early carditis and has adjusted thresholds based on population risk. Typically, ARF diagnosis requires two major, or one major and two minor manifestations, plus evidence of recent Strep A infection. Chorea is an exception, as its occurrence can confirm ARF on its own[2].

Distinguishing a first episode of ARF from a recurrence can be difficult without evidence of a previous episode. In patients with known RHD, diagnostic thresholds for recurrence are lower than in those without a previous documented episode of ARF[2]. Differentiating acute valvulitis from chronic valvulitis may require comparison with previous echocardiograms, when possible. Emerging evidence suggests that the revised Jones criteria have good sensitivity in high-risk populations[92]. A Ugandan follow-up study found that among children who did not meet ARF criteria, only 1.5% developed mild RHD and none progressed to severe disease[92] — maintaining adequate specificity and supporting the reliability of the criteria when applied with appropriate diagnostic and laboratory resources.

However, in many high-burden, resource-limited settings, ARF remains underdiagnosed. In northern Uganda, estimated ARF incidence was 25 per 100,000 among children, despite few clinical diagnoses[3]. In Sudan, 11% of febrile children with initially unknown causes were later found to have carditis on echocardiography[93]. These findings highlight the crucial role of echocardiography in diagnosing

ARF and improving diagnostic accuracy in endemic areas[94]. In high-risk populations in resourced countries such as Australia and New Zealand, participant characteristics, clinical, biochemical and echocardiographic data can be comprehensively phenotyped and international heterogeneity demonstrated[95].

Rheumatic heart disease. Echocardiography remains the gold standard for diagnosis of RHD, offering detailed assessment of the location and severity of valvular lesions as well as secondary complications such as reduced systolic function, pericardial effusion, infective endocarditis and pulmonary hypertension[96]. Electrocardiography (ECG) also has a complementary role, particularly in the detection and monitoring of atrial fibrillation, a frequent complication of advanced RHD.

In 2012, the WHF criteria provided a standardized approach to the echocardiographic diagnosis of RHD, in particular in patients who did not have a history of ARF (up to 90% of those diagnosed with RHD in endemic countries)[97]. A decade of learning prompted a revision in 2023, which introduced several important changes[7] (Boxes 1 and 2). These revisions include two sets of criteria: screening criteria (discussed below) for non-experts in field settings and confirmatory or diagnostic criteria for experts. Additionally, the terms 'borderline', 'latent', 'definite' and 'subclinical' are no longer recommended and are replaced by a new staging system (A–D) that aligns with other cardiac disease classification systems, reflecting the natural heterogeneity of disease and progression risk. Additional updates include weight-based measurements to assess regurgitation jet length and new recommendations for the management of early-stage disease (discussed below, in management)[7].

However, waiting for patients to present with symptomatic RHD in endemic settings is problematic. Late presentation is the most common presentation, often after complications such as heart failure, arrhythmias, infective endocarditis or thromboembolic events have developed[30,98]. Patients who present late have high morbidity and mortality; studies have reported up to 30% mortality within the first year of diagnosis[99].

We now know with some certainty that patients with advanced RHD (WHF stage C or D) represent the tip of the iceberg[100]. In 2016, a pyramidal model for RHD presence and progression was introduced[101,102], and substantial evidence suggest that early RHD (WHF stage A or B) is an intermediate phase that many patients go through between ARF and the development of advanced valvular damage[103]. First, early RHD is predominantly found in populations in which advanced RHD is common, suggesting a shared disease pathway, and not found in regions without advanced RHD (such as low-risk populations in high-income areas)[104,105]. Second, longitudinal studies show that children with early RHD are at risk of clinical progression, including worsening valve lesions and increased morbidity, particularly in the absence of intervention[106]. Finally, a randomized trial has shown that secondary prophylaxis with antibiotics can significantly reduce disease progression[8], indicating that early RHD is not only a precursor to severe disease but also a modifiable stage (Fig. 6). Together, these findings reinforce the concept of early RHD as a crucial target for timely diagnosis and prevention strategies.

RHD screening

The above-mentioned findings prompted the WHO to recommend routine screening for early RHD in its 2024 guidelines[107], moving this approach from research into clinical practice. A two-step, screen and confirm strategy is now advised for children and adolescents 5–19 years of age and for pregnant women in high-risk populations — defined as moderate or high RHD prevalence of >100 per 100,000 population per year[107] (Fig. 7). Drawing on more than a decade of research, the WHO also endorsed the use of hand-held echocardiography devices, which reduce costs and improve portability without drastically compromising diagnostic performance[108]. Effective screening, however, requires a trained workforce, simplified protocols and clear care pathways for those who test positive. Task-shifting to nurses and other health workers is supported by numerous studies and the adoption of standardized screening criteria, using simplified protocols[7]. In the future, artificial intelligence may further streamline screening by offering diagnostic

Box 1 | Staging of RHD by echocardiography according to the 2023 WHF guidelines[7]

Stage A: early RHD; minimal echocardiographic criteria for RHD

- Applies only to individuals ≤20 years of age
- Clinical risk: might be at risk of valvular heart disease progression
- Echocardiographic features (fulfilling the confirmatory criteria for pathological regurgitation of the 2023 World Heart Federation (WHF) guidelines): the presence of mild mitral or aortic regurgitation without morphological features

Stage B: mild RHD

- Can apply to any age
- Clinical risk: at moderate or high risk of progression and at risk of developing symptoms of valvular heart disease
- Echocardiographic features: evidence of mild valvular regurgitation and at least one morphological feature in individuals ≤20 years of age and at least two morphological features in individuals >20 years of age; or mild regurgitation in both mitral and aortic valves

Stage C: advanced RHD at risk of complications

- Can apply to any age
- Clinical risk: at high risk of developing clinical complications that require medical or surgical intervention
- Echocardiographic features: moderate or severe mitral or aortic regurgitation, any mitral or aortic stenosis, pulmonary hypertension or decreased left ventricular systolic function

Stage D: advanced RHD with clinical complications

- Can apply to any age
- Clinical risk: established clinical complications include cardiac surgery, heart failure, arrhythmia, stroke and infective endocarditis
- Echocardiographic features: moderate or severe mitral regurgitation, moderate or severe aortic regurgitation, any mitral or aortic stenosis, pulmonary hypertension and decreased left ventricular systolic function

RHD, rheumatic heart disease. Adapted from ref. 7, Springer Nature Limited.

Box 2 | Confirmatory echocardiography criteria for pathological valve dysfunction from the 2023 WHF guidelines

Pathological (at least mild) mitral regurgitation (all criteria must be met)

- Observed in at least two views
- Observed in at least one view, mitral regurgitation jet length ≥1.5 cm (in individuals weighing ≤30 kg) or ≥2.0 cm (in individuals weighing ≥30 kg)
- Velocity ≥3.0 m/s for one complete envelope
- Pan-systolic jet in at least one envelope

Pathological (at least mild) aortic regurgitation (all criteria must be met)

- Observed in at least two views
- Observed in at least one view, aortic regurgitation jet length ≥1.0 cm
- Velocity ≥3.0 m/s in early diastole
- Pan-diastolic jet in at least one envelope

WHF, World Heart Federation. Adapted from ref. 7, Springer Nature Limited.

support and easing the training burden, paving the way for broader implementation. Registry-based care is recommended (the second edition of the WHO non-communicable diseases best-buys[109]) as well as ensuring an adequate supply of antibiotics for secondary prophylaxis.

Implementing screening programmes at scale requires multisector partnerships and government engagement. One example is the ADUNU (the Acholi word for heart) programme in northern Uganda[110], a partnership between the Rheumatic Heart Disease Research Collaborative in Uganda (RRCU) and the Uganda Ministry of Health, focusing on integrating screening into the public health system. Similarly, an implementation science study of task sharing echocardiography screening in Australian First Nations communities will provide health system strengthening insights relevant to similar HICs where RHD remains prevalent in marginalized populations[111]. Lessons learned from these programmes could provide a template for scale-up in other countries.

Prevention of ARF and RHD

Multiple opportunities exist in the path between Strep A exposure and RHD to intervene and prevent disease progression (Fig. 8). In 2024, the WHO published guidelines on the prevention, screening and diagnosis of ARF and RHD[107] (Table 2).

Primordial prevention. Primordial prevention focuses on addressing the social determinants of health that increase the risk of ARF and RHD. Historical declines in ARF in HICs occurred before the widespread use of antibiotics, coinciding with improvements in housing and socioeconomic conditions[112]. Key risk factors — overcrowded and poor housing infrastructure, inadequate sanitation, limited access to healthcare, low income and low health literacy — increase exposure to Strep A[113] (Fig. 3).

In addition, studies have reinforced the association between ARF and several social and environmental conditions, including household crowding (OR 3.88), poor access to healthcare (OR 2.07), skin infections (adjusted OR (aOR) 2.53) and sore throat (aOR 2.33)[32]. Improvement in living conditions, sanitation infrastructure (for example, access to clean water, soap and sewage systems) and reduction in overcrowding are proven strategies to reduce Strep A transmission.

There is growing momentum to prioritize RHD primordial prevention, and latest findings have outlined key research priorities[114]. A small, community-led initiative in northern Australia has shown early promise — community health workers, working directly with families, supported household interventions and healthcare navigation. Although this small study did not reach statistical significance, it demonstrated encouraging results that such interventions may contribute to reducing Strep A infections and ARF incidence[115]. Although there may be challenges in scaling such initiatives, particularly in LMICs owing to the intricacies of the socioeconomic determinants of health, further investment in determining how health system strengthening may incorporate strategies to address primordial risk factors is vital. An integrated universal healthcare approach that emphasizes health systems strengthening and equity is recommended.

Strep A vaccine. Global efforts to develop a Strep A vaccine have accelerated over the past decade. In 2019, the WHO published a research and development roadmap and the preferred product characteristics to guide vaccine design[116]. Major initiatives — including the Australian STREP A Vaccine Initiative (ASAVI) and the Strep A Vaccine Global Consortium (SAVAC) — have been established to advocate that safe, effective and affordable vaccines reach populations most in need. To support vaccine advocacy and testing, sentinel surveillance sites have been set up in The Gambia, Malawi, Fiji and India to better understand the global burden of Strep A and prepare for clinical trials. Large international research networks have been formed, such as the iSPY consortium, working to characterize natural immunity to Strep A, and Wellcome Trust-funded networks working on immune correlates of protection, which are key to evaluating vaccine efficacy. The successful development of a human challenge model further offers a valuable tool to accelerate candidate testing[117].

Although early trials are focused on prevention of pharyngitis, there is a pressing need to evaluate severe outcomes — particularly, ARF and RHD as primary end points, which is essential to ensure future vaccines are suitable and prioritized for LMICs where the burden of non-suppurative Strep A-mediated disease is greatest.

Primary prevention. Primary prevention aims to treat superficial Strep A infections — primarily pharyngitis and impetigo — with antibiotics before the occurrence of abnormal immune response that causes ARF[114]. This approach depends on individuals recognizing symptoms and accessing healthcare. However, most Strep A infections are mild, asymptomatic and go unrecognized, limiting timely diagnosis and treatment. One systematic review evaluating the diagnostic performance of clinical prediction rules in detecting Strep A infections in people presenting with pharyngitis suggested very low certainty of evidence[118].

Evidence shows that treating Strep A pharyngitis reduces ARF incidence[119,120]. Historical data from the US military in the 1940s demonstrated that antibiotic treatment lowered ARF incidence among recruits[121]. Additionally, a large school-based programme in New Zealand involving 22,000 students reported a 28% reduction in ARF, although results were not statistically significant ($P = 0.27$)[122]. Penicillin remains an effective treatment for Strep A infections in LMICs[123], with a 'treat all' approach found to be the least costly strategy for primary prevention of ARF[124].

Strep A skin infections (impetigo) may also trigger ARF. This is particularly relevant for Australian First Nations communities, where ARF rates are high, pharyngitis is relatively uncommon (4 per 100 per year) but impetigo is widespread (44.5%)[125,126]. Similar findings in New Zealand have linked ARF episodes to recent skin infections[127,128]. A trial using a trimodal strategy to reduce Strep A skin infections achieved a sustained decline in infection rates, although a direct causal link to intervention stages was not established[129]. The contemporary understanding of RHD progression — many individuals with RHD report no history of ARF and most ARF cases lack a clear link to a superficial Strep A infection (and occur in settings that lack Strep A diagnostic testing) seems to complicate the traditional primary prevention paradigm. Regardless, early clinical recognition and prompt diagnostic testing (if available) for superficial Strep A infection, regardless of the anatomical site of the infection, followed by treatment with antibiotics (penicillin) remains an important pillar of prevention for ARF and subsequent RHD.

Secondary prevention. Once ARF has occurred or RHD is established, secondary antibiotic prophylaxis is the cornerstone of RHD management, aiming to prevent recurrent ARF and progression of valvular damage[107]. Historical data showed that 40–60% of ARF cases progressed to RHD[130]. One Australian study found that 35% of children with ARF developed RHD within 1 year and 61% within 10 years. Secondary antibiotic prophylaxis is also associated with regression of mitral regurgitation (a marker of rheumatic carditis), from 20% to 70% in some cases[131].

A landmark randomized controlled trial (RCT) to compare secondary antibiotic prophylaxis with no treatment in children with early, screen-detected RHD showed significantly less disease progression in the secondary antibiotic prophylaxis group over 2 years (0.8% versus 8.2%)[8]. These findings support echocardiographic screening and early intervention, particularly in LMICs, where late presentation with complications such as heart failure and stroke is common (see 'Diagnosis' section).

Intramuscular benzathine benzylpenicillin (BPG) is still the preferred mode of secondary antibiotic prophylaxis[107]. Studies have demonstrated that intramuscular penicillin is more effective than oral penicillin; however, these were in historical cohorts with different preparations of oral penicillin compared with current formulations[132]. A non-inferiority trial that may challenge this historical understanding is ongoing[133]. A target serum penicillin level of >20 ng/ml (performed only for pharmacokinetic studies) is recommended to prevent recurrent Strep A infection[134]. Higher BPG doses can be well tolerated and may further reduce recurrences, although maintaining therapeutic levels over the standard 4-week interval is challenging[135]. The recommended duration of secondary antibiotic prophylaxis is 10 years or until the age of 21 years, or lifelong secondary antibiotic prophylaxis in patients with severe RHD. A dose is administered every 28 days and for some patients with ARF recurrence despite adherence, 3-weekly dosing may be beneficial[132,136].

Allergy and anaphylaxis rates to BPG remain low, estimated at 0.25%[137]. However, some studies have reported adverse outcomes including cardiac arrest and sudden cardiac death after intramuscular BPG injection[138,139]. Anecdotal evidence suggests that these outcomes are a result of cardiovascular collapse and sudden death among patients with severe valve disease. These patients tend to have low cardiovascular reserve and dehydration. An American Heart Association advisory now recommends that patients with severe valve disease should receive oral penicillin instead of intramuscular BPG[140]. If a decision to administer intramuscular BPG is made, the patient should be pre-hydrated, cardiac observations should be taken, connected to a cardiac monitor if available and have equipment for cardiorespiratory resuscitation readily available[140]. Strategies for pain mitigation and avoidance of vasovagal syncope, including administration of BPG in the prone position, should be taken into consideration.

Adherence to intramuscular BPG is often limited by pain both during and after the injection and the availability of the drug in resource-limited settings. Evidence-based strategies to improve patient experience include allowing preferred injection sites, distraction, local

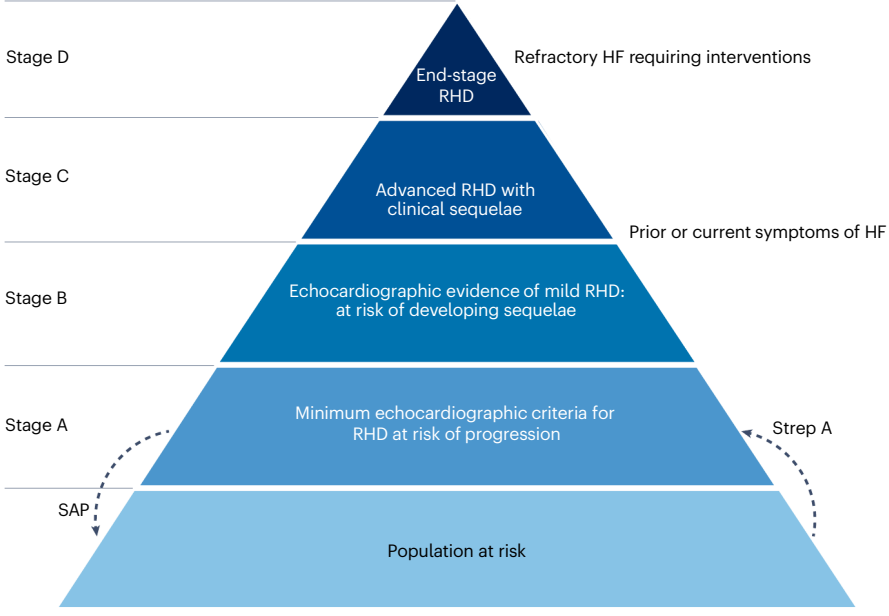


Fig. 6 | Revised pathogenic pathway of early to advanced RHD. A revised pyramid takes into account the World Heart Federation (WHF) stages as per the WHF revised guidelines and links them to heart failure (HF) stages. The arrow on the left reflects the evidence that secondary antibiotic prophylaxis (SAP) can result in regression or normalization of stage A WHF criteria and move individuals to population at risk. The arrow on the right reflects the fact that exposure to group A *Streptococcus* (Strep A) moves individuals from being population at risk to at risk for further progression. RHD, rheumatic heart disease.

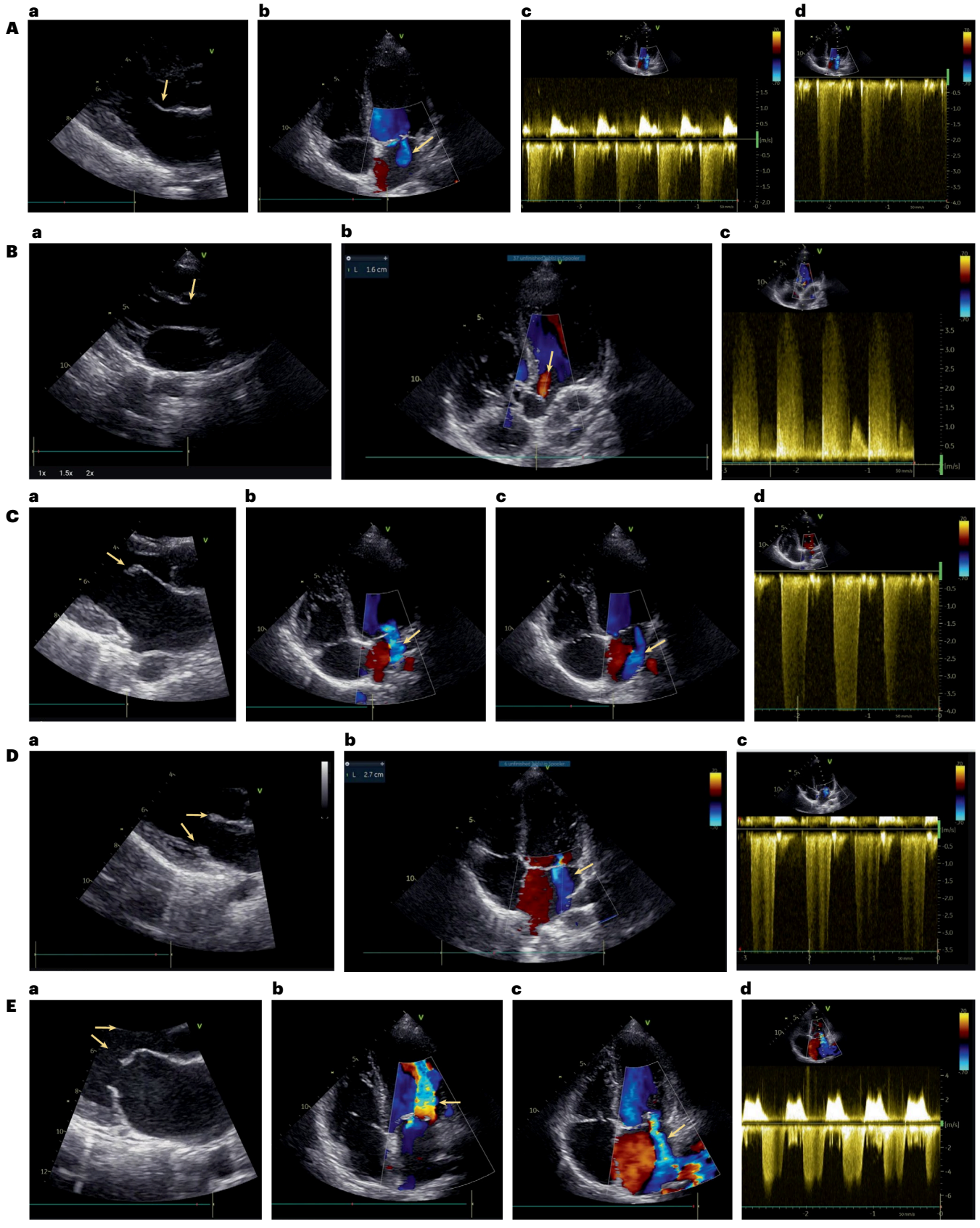

A
a
b
c
d
B
a
b
c
1x 1.5x 2x
1.6 cm
C
a
b
c
d
D
a
b
c
2.7 cm
E
a
b
c
d

Fig. 7 | Echocardiographic images of RHD stages A–C. These echocardiograms were taken from a school screening programme, which included the conduct of confirmatory studies in the field (school setting) in Uganda, 2024. **A**, Rheumatic heart disease (RHD) stage A. **Aa**, Normal morphology of the mitral valve. The yellow arrow shows a thin anterior mitral valve leaflet. **Ab**, Mitral regurgitation (MR) jet (yellow arrow) that is 2.2 cm long. **Ac**, Spectral Doppler showing pan-systolic MR. **Ad**, Spectral Doppler showing MR velocity >3.0 m/s. **B**, RHD stage A. **Ba**, Normal morphology of the aortic valve. The yellow arrow shows a thin right coronary cusp. **Bb**, Aortic regurgitation (AR) jet (yellow arrow) that is 1.5 cm long. **Bc**, Spectral Doppler showing pan-diastolic AR jet with velocity >3.0 m/s. **C**, RHD stage B. **Ca**, Morphological features of the mitral valve (MV) including anterior leaflet thickening (yellow arrow). **Cb**, Mosaic MR jet (yellow arrow) that is 2.1 cm long. **Cc**, A longer MR jet (yellow arrow) measuring 2.7 cm. **Cd**, Spectral Doppler showing pan-systolic MR with velocity >3.0 m/s. **D**, RHD stage B. **Da**, Morphological features of the mitral valve including anterior and posterior leaflet thickening (yellow arrows). **Db**, MR jet (yellow arrow) that is 2.7 cm long. **Dc**, Spectral Doppler showing pan-systolic MR with velocity >3.0 m/s. **E**, RHD stage C. **Ea**, Morphological features of the MV including anterior and posterior leaflet thickening (yellow arrows) and mobility abnormalities. **Eb**, Acceleration of forward flow (yellow arrow) through the mitral valve, suggesting obstruction at the valve level. **Ec**, Eccentric MR jet (yellow arrow) that is 3.8 cm long. **Ed**, Spectral Doppler showing pan-systolic MR with velocity >3.0 m/s. Electrocardiogram gating was not used.

anaesthetic, nitrous oxide and clonidine[141]. A community health worker strategy based on active case management and peer groups has achieved high adherence (98% versus 80% in usual care) in Uganda[142].

In addition, alternative BPG delivery methods are being explored. An Australian team has developed a long-acting subcutaneous BPG infusion; preliminary studies suggest that subcutaneous infusion is well tolerated, minimally painful and achieves comparable penicillin levels to 4-weekly intramuscular injections[143].

Management

Acute rheumatic fever

Currently, the management of ARF consists only of symptom control and prevention of further Strep A infections as no disease-modifying treatment has yet been shown to be effective. Symptom control includes the use of anti-inflammatory therapy for arthritis, heart failure management for advanced carditis and anticonvulsant medication for severe chorea[144]. Secondary antibiotic prophylaxis is part of ARF management to prevent further Strep A infection and, therefore, recurrent ARF (Box 3).

Several studies of steroid therapy to reduce the severity of rheumatic carditis have been conducted; however, results have not been definitive[145]. An ongoing RCT of steroids in chorea in Australia and New Zealand may inform future guidance[146].

ARF diagnoses may be made at a clinic or district level facility, if echocardiography is available. The use of task-sharing and hand-held echocardiography with telecardiology consultations followed by scheduled assessment by specialists should be considered in resource-limited and remote settings[107]. Transfer to a district or tertiary hospital is recommended for optimal medical management in those with considerable valvular involvement. Secondary antibiotic prophylaxis should be started during the admission with counselling and education for the patient and their caregivers. Importantly, notification of the episode to health authorities is mandatory in certain countries, which allows for the commencement of registry-based follow-up and care. Registry-based care is recommended (a 2023 WHO non-communicable disease best-buy[109]) as well as ensuring an adequate supply of antibiotics for secondary prophylaxis. There is an urgent need for modern therapeutics discovery, testing and repurposing[147]. Indeed, the launch of an adaptive platform trial in New Zealand and Australia marks a crucial first step — pioneering the investigation of potential treatments in the contemporary era.

Rheumatic heart disease

Managing patients with RHD is complex, expensive and requires strong national health systems in countries where, often, the infrastructure for comprehensive management is limited. Treatment options for severe disease are primarily focused on managing comorbidities and complications, which consist of medical management of heart failure or arrhythmias, and anticoagulation as catheter and surgical therapies continue to be lacking in low-resource settings[107].

Early and mild RHD. Consensus is growing regarding the treatment of early RHD (WHF stage A or B), diagnosed on echocardiography, in an asymptomatic population (Boxes 1 and 2). A landmark RCT comparing

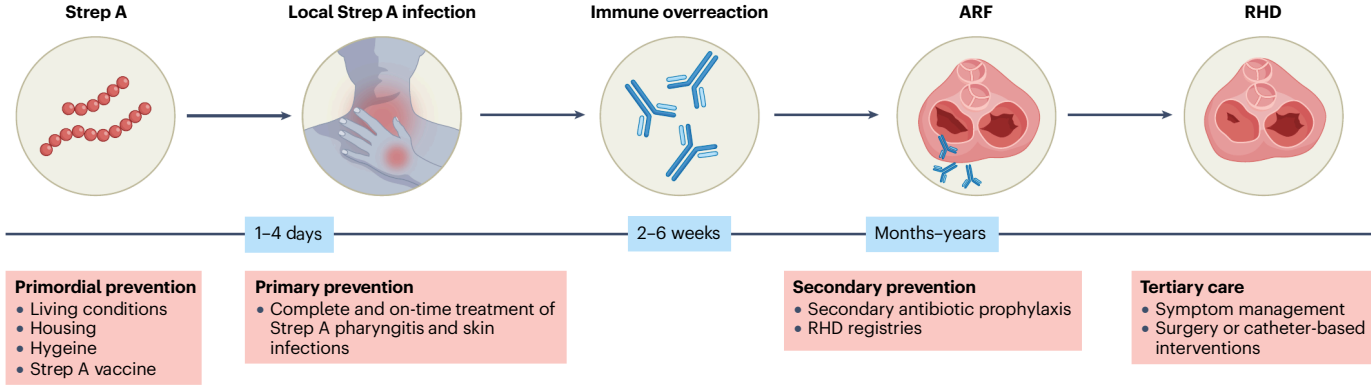


Fig. 8 | Opportunities to prevent progression and adverse outcomes in the progression of Strep A to RHD. This figure aligns interventions at primordial, primary, secondary and tertiary levels in the pathogenic pathway from group A *Streptococcus* (Strep A) to rheumatic heart disease (RHD). It delineates the time frames (1–4 days from exposure to Strep A to pharyngitis), 2–6 weeks from this exposure to acute rheumatic fever (ARF) and then months to years to develop and manifest RHD. Opportunities for intervention along the pathways are presented. Adapted from slides produced by T. Agenson and J. Katzenellenbogen (2018).

Table 2 | 2024 WHO guidelines on RHD diagnosis, screening and prevention

Stage	Strength of recommendation	Certainty of evidence
Diagnosis and screening		
The Jones criteria should be used for ARF diagnosis in children, adolescents and adults with suspected RF	Strong	Low
Among children, adolescents and adults with suspected ARF or RHD in settings where standard echocardiography is not available, HHE can be used for diagnosis of ARF-carditis and RHD	Strong	Very low for ARF-carditis, moderate for RHD
In populations or settings with moderate to high risk of RHD, standard echocardiographic or HHE can be considered, to improve early detection of RHD among pregnant women during antenatal care	Conditional	Very low
In populations with moderate to high RHD prevalence, echocardiographic screening using standard echocardiography or HHE can be implemented for early detection of RHD among children and adolescents 5–19 years of age.	Strong	High
Treatment		
Children, adolescents and adults diagnosed with ARF or RHD should be prescribed antibiotic prophylaxis to prevent ARF recurrence	Strong	Moderate
Antibiotic prophylaxis should be prescribed for children and adolescents found to meet minimum criteria for RHD on echocardiography screening to prevent disease progression	Strong	Moderate
Antibiotic prophylaxis can be prescribed for adults 20 years of age and older who are found to meet minimum criteria for RHD on echocardiography screening	Conditional	Very low
Antibiotic prophylaxis should be given to children and adolescents who have advanced RHD to prevent RF recurrence	Strong	Very low
Antibiotic prophylaxis can be given to adults 20 years of age and older who have advanced RHD to prevent ARF recurrence based on shared decision-making between the patient and the treating healthcare provider	Conditional	Very low
Intramuscular BPG is the preferred first-line approach to prevent recurrence of RF in patients with prior ARF or RHD	Strong	Moderate
Prevention		
Recommended that health workers provide evidence-based education focused on the relationship between infections of the pharynx and skin potentially caused by Strep A and ARF/RHD, and thus the importance of treating these infections appropriately, particularly in moderate- to high-risk settings or populations	Conditional	Very low
Children, adolescents and adults with sore throat and a positive diagnostic test (either POC testing or microbiological confirmation) for Strep A pharyngitis should be treated with antibiotics to prevent ARF and RHD	Strong	Moderate
In populations at moderate to high risk of ARF and RHD and where diagnostic testing to confirm Strep A (with either POC testing or microbiological confirmation) is not available, children and adolescents with clinically suspected Strep A pharyngitis should be treated with antibiotics to prevent ARF and RHD	Strong	Very low
For patients with a positive diagnostic test for Strep A pharyngitis or with clinically suspected Strep A, penicillin (intramuscular or oral) is recommended as first-line treatment for the prevention of ARF and RHD	Conditional	Low

For more information, see ref. 107. ARF, acute rheumatic fever; BPG, benzathine benzylpenicillin, HHE, hand-held echocardiography; POC, point of care; RF, rheumatic fever; RHD, rheumatic heart disease; Strep A, group A *Streptococcus*.

secondary antibiotic prophylaxis with no treatment in children with early, screen-detected RHD showed significantly less disease progression in the antibiotic prophylaxis group over 2 years (0.8% versus 8.2%)[8]. A similar conclusion was reached in the Nepalese cluster randomized trial[148]. These findings support echocardiographic screening and early intervention, particularly in LMICs, where late presentation with complications such as heart failure and stroke is common (see 'Diagnosis' section). Secondary antibiotic prophylaxis should be implemented in endemic regions to retard disease progression, reverse subclinical valvular lesions and control transmission from infectious reservoirs.

Clinical RHD. The predominant global burden of symptomatic RHD is concentrated in areas with restricted or no access to surgical or catheter-based interventions. Yet, according to established data, surgical or catheter-based procedures are the guideline-directed treatment for severe or symptomatic RHD[149]. Although there is less evidence to demonstrate the impact of pharmacological treatment on disease outcome in endemic regions, medical care frequently represents the sole alternative and may facilitate symptomatic enhancement.

Medical treatment. In patients with severe valvular regurgitation presenting with heart failure and severe heart chamber dilatation, diuretics (especially loop diuretics), digoxin, β-blockers, non-dihydropyridine calcium channel blockers and angiotensin-converting enzyme (ACE) inhibitors have been associated with improved quality of life, especially in symptomatic patients[150]. This strategy helps to stabilize the patient while preparing or waiting for the definitive management using surgical interventions. Given that patients present with multi-valve, mixed-valve lesions and concomitant complications, careful consideration of appropriate pharmacological therapy is required[151].

In patients with severe mitral stenosis with or without atrial fibrillation, vitamin K antagonists have been shown to substantially reduce the risk of major adverse cardiovascular events and the risk of cardiovascular death without notably increasing the risk of bleeding. Vitamin K antagonists were shown to be superior over the newer

non-vitamin K antagonist (rivaroxaban) in the INVICTUS clinical trial[152]. Hence, vitamin K antagonists remain a guideline recommendation as the anticoagulant of choice in rheumatic mitral stenosis and atrial fibrillation.

Interventional procedures. Catheterization-based interventional procedures are important modalities of treatment in the management of rheumatic mitral stenosis owing to their cost effectiveness as well as a rapid recovery time. Although safe and usually very relevant in patients who may not be immediate candidates for surgery, catheterization-based interventional procedures are not easily available in endemic regions where severe mitral valve stenosis is a common feature of RHD. Percutaneous mitral commissurotomy (PMC) is the main intervention of choice in patients with severe mitral stenosis who meet criteria per the Wilkins score[153]. From the 2020 American College of Cardiology and American Heart Association (ACC–AHA) guidelines[154] and the 2021 European Society of Cardiology and European Association for Cardio-thoracic Surgery (ESC–EACTS) guidelines[155], PMC is the most suitable treatment when the patient has severe mitral stenosis (valve area ≤1.5 cm^2), favourable valve morphology, absence of moderate or severe mitral regurgitation, left atrial thrombus or severe valvular calcification. PMC has a good safety profile even in patients with suboptimal conditions (Wilkins score 7–8) leading to a considerable improvement in clinical symptoms, quality of life and in addition delaying the need for surgical correction. Studies have shown that the degree of postoperative change in mitral valve area can serve as a key indicator of long-term success and risk reduction in PMC[156]. Patients with a mitral valve area increase of >50% showed a significantly reduced risk of recurrence[156]. Although complications (such as tamponade or valve leaflet rupture) are rare (2–5%), on-site surgical back-up is typically still required. Long-term benefit after balloon mitral valvuloplasty is observed in ~50% of patients after 10 years[157].

Surgical treatment. Surgical interventions for RHD have been shown to be substantially associated with improvement in the quality of life of patients with severe RHD; however, the stark reality is that the predominant global burden of severe RHD is concentrated in regions with little or no access to surgical or catheter-based interventions[19].

A Ugandan series showed that without surgery, 24% of patients with RHD died at 1 year[158]. With surgical intervention, mortality was markedly reduced[158]. Surgical correction of mitral valvular stenosis is associated with notably better outcomes with reduced mortality compared with medication alone and is the recommended mode of treatment when the patient's native valve condition is insufficient to support PMC for mitral stenosis and regurgitant aortic and mitral lesions[96]. In patients with mitral valve disease, the choice of either to repair or to replace the valve is an ongoing debate. A meta-analysis involving more than 19 studies with a patient pool of ~10,724 has shown significantly lower early mortality rates and improved long-term survival in patients who underwent mitral valve repair compared with those who had mitral valve replacement[159]. Furthermore, mitral valve repair is also associated with significantly reduced risk of valve-related complications. However, mitral valve repair is associated with a higher likelihood of repeat surgeries than mitral valve replacement and required a high level of surgical expertise[159].

The choice between bioprosthetic valve and mechanical valve for mitral valve replacement should be discussed with the patient and family, with considerations given to age, sex, availability and cost. Mechanical valves are associated with superior long-term outcomes, reduced rates of all-cause mortality and re-operation compared with bioprosthetic valves especially in patients <65 years of age[160]. However, in resource-constrained environments such as sub-Saharan Africa, bioprosthetic valves may be associated with better outcomes than mechanical valves with suboptimal international normalized ratio monitoring. One common complication is valve thrombosis as a result of low resources for international normalized ratio monitoring, low health literacy and poor concordance with anticoagulation protocols for mechanical valve care[161].

Treatment of RHD in pregnancy

RHD during pregnancy is challenging to manage, mostly owing to the pregnancy-induced alterations in cardiac output, the hypercoagulable state that increases the risk of thrombus formation and the concomitant reductions in vascular resistance observed in the middle

Box 3 | Management of ARF

Diagnosis
Admission to hospital; investigations to confirm acute rheumatic fever and exclude other pathologies. Blood tests including acute phase reactants and serology for the streptococcal organism; electrocardiography; echocardiographic evaluation is key.

Eradication of group A *Streptococcus*
Oral penicillin V potassium (penicillin VK) for 10 days or single dose of intramuscular benzathine penicillin G (preferred).

Arthritis and/or arthralgia and symptomatic treatment
Paracetamol until the diagnosis has been confirmed; NSAIDs after confirmation of the diagnosis. Corticosteroids may be considered in intractable cases for which NSAIDs cannot be used. Use of corticosteroids: conditional recommendation, low certainty evidence.

Carditis or heart failure
Bed rest, fluid restriction, heart failure medications (furosemide, spironolactone, angiotensin-converting-enzyme inhibitor); surgery for intractable heart failure associated with severe mitral or aortic regurgitation; preferable to defer surgery until acute rheumatic activity has resolved.

Chorea
Haloperidol or carbamazepine can be considered if the abnormal movements interfere with daily activities; valproic acid reserved for refractory cases; multidisciplinary input as required for significant motor or neuropsychiatric manifestations.

Discharge procedure
Consider discharge once there is clinical improvement and/or reduction in erythrocyte sedimentation rate or C-reactive protein; notification to health authorities; patient and family education; secondary prophylaxis; outpatient follow-up.

For more information, see ref. 134. ARF, acute rheumatic fever.

Box 4 | Patient experience

Living with rheumatic heart disease (RHD) for over two decades has shown me that RHD is far more than just a medical diagnosis. At age 11, a series of poorly treated sore throats led to my diagnosis and two open-heart surgeries. I had no idea how profoundly this condition would affect my life as a girl growing up in a low-resource environment. From childhood to adulthood, RHD has subtly but persistently affected my daily functioning, mental and emotional well-being, education, career, social life, financial stability and access to healthcare. It has posed challenges to my quality of life in both visible and invisible ways.

Physically, symptoms such as arrhythmia, fatigue and joint pain limited my ability to engage in 'normal' activities, leading to dependency and low self-esteem. Repeatedly missing school, work and social events often left me isolated, with a fear of missing out. Emotionally, I experienced fear, uncertainty about future living with RHD, and what might be described as medical trauma. Even after years, I sometimes get anticipatory anxiety when familiar symptoms resurface, reminding me of medical crises and surgeries. Yet through this, I have built resilience and self-awareness.

Sometimes it distresses me as a woman because stigma reduces me to my diagnosis and makes people doubt my capacity to work, get married or live a full life. These moments remind me how little is known about RHD and how many women can live fulfilling lives and follow their own paths that are right for them. My education and job opportunities were interrupted by illness. I once lost a job because of a protracted recuperation. Financially, I and my family were under a great deal of stress owing to the cost of treatment requirements. The emotional and physical stresses of accessing medicine and the financial burden related to RHD have led many to feel that enduring pain is better than burdening their families, affecting both self-esteem and medication adherence.

RHD changed my life, but it did not take away my voice. I share my story to raise awareness, fight stigma and advocate for a future where no one suffers needlessly from this preventable disease.

and late trimesters. Poorly treated RHD or severe RHD can result in adverse maternal and fetal outcomes, such as heart failure, pulmonary hypertension, arrhythmias, thromboembolism and pregnancy-related mortality[162]. One study demonstrated balloon mitral valvotomy to relieve severe mitral stenosis as a rescue strategy in symptomatic pregnant women[163].

Given the possibility of pregnancy in patients with RHD, mitral valve repair is a more suitable treatment modality than mitral valve replacement as it obviates the need for long-term anticoagulant therapy. Vitamin K antagonists such as warfarin are associated with an increased risk of fetal anomalies, miscarriage and stillbirth[164]. The option of low-molecular-weight heparin used for anticoagulation in pregnancy has its own challenges, such as the need for close monitoring of anti-Xa levels and its low efficacy in preventing thrombus formation on mechanical valves compared with warfarin[165]. A multidisciplinary cardio-obstetrics team approach has become vital to provide the appropriate level of care during pregnancy[166].

Management of complications in RHD

A common complication of RHD especially in patients with rheumatic mitral stenosis is atrial fibrillation[167]. Usually, the management of atrial fibrillation in patients with RHD is aimed at symptom control and the prevention of cardiovascular events such as stroke and embolism using anticoagulation and appropriate valve interventions[168]. The CHA-DAS2-VASc score for atrial fibrillation stroke risk uses an acronym for risk factors: congestive heart failure (1 point), hypertension (1 point), age 75+ years (2 points), diabetes (1 point), stroke/transient ischaemic attack (TIA) (2 points), vascular disease (1 point), age 65–74 years (1 point) and sex (female sex, 1 point). Scores help to guide anticoagulation, with higher scores indicating greater stroke risk and stronger recommendations for blood thinners, although scores of 0 (male patients) or 1 (female patients) generally do not need treatment, whereas scores of ≥2 (male patients) or ≥3 (female patients) typically require anticoagulation. Although this score helps in the assessment of the risk of stroke in such patients, thereby guiding the decision for anticoagulation, patients with RHD are generally much younger than individuals in the atrial fibrillation cohorts upon which these scores were based[5,152,169]. Atrial fibrillation and multi-valve disease usually show markedly increased mortality when treated with medication alone and therefore, other interventions are typically necessary[169].

Infective endocarditis, an infection of the endocardial surface of the heart, frequently occurs in patients with RHD, particularly among those with previous valve dysfunction, younger patients, those with prosthetic valves and other comorbidities[170]. As a high-risk group, these individuals require antibiotic prophylaxis before any surgical manipulation or operation involving the bowel, respiratory and urological organs. Although mortality from infective endocarditis can be as high as 25–50% even in the best of centres, surgical removal of large vegetations may be required in addition to antibiotics for the treatment of established infective endocarditis[171].

Quality of life

RHD profoundly affects health-related quality of life (HRQoL), influencing physical, emotional and social well-being across the lifespan. Several disease-specific and generic instruments have been used to evaluate HRQoL among patients with RHD. In paediatric populations, the Paediatric Quality of Life Inventory (PedsQL) Cardiac Module is frequently used, to assess physical symptoms, treatment anxiety, cognitive challenges and communication issues[172]. Among adults, studies often use the EQ-5D and the SF-36 tools, which measure broader domains such as physical functioning, pain, vitality and emotional well-being[150,173]. Disease-specific adaptations, such as incorporating questions around anticoagulation therapy after valve replacement, have been crucial to capturing the nuanced experiences of patients with RHD[174].

Overall, evidence demonstrates that people with RHD experience considerably impaired HRQoL. In one study from India, >50% of patients with ARF or RHD reported moderate to severe impairments in at least one domain of HRQoL[175]. Although some studies found that HRQoL scores improved after diagnosis and treatment initiation, these scores still remained lower than the general population averages, particularly in domains related to mobility and anxiety[176]. In Uganda, children who underwent valve replacement surgery experienced improvements in HRQoL scores; however, a large proportion still reported concerns about physical activity limitations and adverse effects related to treatment[172]. In Namibia, patients' HRQoL scores improved after diagnosis and treatment initiation, but remained lower than general population norms, particularly in domains related to mobility and

anxiety[173]. In comparison with other chronic cardiovascular conditions, such as congenital heart disease or cardiomyopathy, patients with RHD often exhibit comparable or worse HRQoL scores. Unlike congenital heart diseases, which typically present in infancy and early childhood and for which established care pathways are widely available in high-income settings, RHD affects children and young adults in resource-limited environments where access to early intervention and ongoing care is limited[175]. The socioeconomic impact, stemming from reduced school attendance or work absenteeism, high out-of-pocket cost of medicines and travel to health facilities, as well as poor access to RHD diagnostics and medications, seem more profound in RHD than in non-infectious cardiac diseases[177].

RHD is also associated with a substantial reduction in life expectancy in endemic settings where surgical and long-term medical care are scarce. Although direct lifespan data were less emphasized in the reviewed studies, cumulative evidence underscores early mortality as a consistent outcome in endemic regions[150,178]. Early identification through screening may offer a pathway to improve both lifespan and quality of life by facilitating timely intervention[178].

Treatment of RHD, particularly surgical valve replacement, markedly improves HRQoL but introduces new challenges. Concerns regarding anticoagulation management, bleeding risk and fear of re-operation persisted among patients in resource-limited settings[172,174]. Similar reports suggest that although Ugandan children experienced better HRQoL after valve surgery, anxiety around lifelong anticoagulation therapy and activity restrictions remained prominent[42]. Notably, access to high-quality surgical care and postoperative follow-up are key determinants of the degree of HRQoL improvement achieved.

In addition to clinical and functional limitations, RHD also exerts a profound psychosocial burden, with depression, anxiety and reduced social participation being common among people with RHD[150]. Children and adolescents with RHD exhibited reduced exercise tolerance, diminished pulmonary function and impaired respiratory muscle strength, all contributing to limitations in physical activity and poor overall quality[179]. Feelings of social stigma, fear of sudden cardiac events and dependence on family members for care compound the psychological toll.

Outlook

In LMICs, ARF and RHD continue to challenge collective efforts to understand[180] and manage the disease and several issues relating to disease biology and epidemiology remain unresolved[181]. However, since the mid-1990s, the incidence of ARF and RHD in HICs has markedly declined largely owing to improvements in living standards, primary care and antibiotic availability. Given this disparity, substantial efforts to understand the disease pathogenesis and effectively implement existing knowledge for the benefit of the large numbers of affected patients continue[114,182,183].

The primary obstacles to development of successful population-based disease control methods for RHD include the scale and heterogeneity of affected populations, the necessity for lifelong monitoring and the recurrent requirement for costly tertiary care. A robust health infrastructure is needed for primary and secondary prevention and

Box 5 | Gaps in preclinical, epidemiological, clinical and implementation research for RHD

Gaps in preclinical and epidemiological research

- More research on group A *Streptococcus* and the development of multiple vaccine candidates are urgently needed
- There are more than 200 group A streptococcal *emm* types, with no consistent associations. Novel ways of typing group A *Streptococcus* are needed
- Further understanding of the role of group A *Streptococcus* skin infections in affected regions
- Although the relationship between living standards and rheumatic fever is clear, the roles of overcrowding, nutrition, education and indoor pollution need clarification
- Understanding of the potential impact of modifying primordial risk factors
- The precise pathophysiological progression of autoinflammation events that culminate in acute injury to heart valves is unclear and could be a therapeutic target
- A diagnostic biomarker might reduce dependence on set diagnostic criteria, as a single definitive laboratory test for rheumatic fever
- Disease-modifying agents in acute rheumatic fever
- The genetic basis for individual susceptibility to rheumatic fever remains unclear

Gaps in clinical and implementation research

- The progression to clinically manifest rheumatic heart disease (RHD) is unclear. Large prospective studies could unravel the natural history of early RHD and the effect of secondary prophylaxis with penicillin on disease progression from early to severe RHD
- Optimal management of mitral and aortic regurgitation and timing of surgery is unclear
- Penicillin injections are not globally accessible. Oral, subcutaneous and long-acting preparations of penicillin need to be tested in larger and diverse populations to address compliance and fear of fatal reactions
- Registries can enable better tracking of patients and greater treatment adherence. Existing registries need to be scaled up, and a global registry should be established
- Innovations in digital data capture have the potential to considerably improve inclusion of patients from diverse settings
- Implementation science approaches to echocardiographic screening require investigation
- Low-cost heart valves under development might eliminate the need for lifelong anticoagulant therapy. Cardiac centres and trained surgeons are needed in affected countries
- Capacity for balloon mitral valvotomy must be increased in catheterization laboratories in low-resource settings
- Pregnancy in women with rheumatic valve disease can be high risk and thus an important cause of maternal mortality. Screening for RHD should be routine in antenatal health checks in affected regions
- Validation studies of the 2023 WHF criteria will demonstrate sensitivity and specificity

disease registries, especially in regions with high disease burden[184]. Cardiologists and surgeons should advocate for improvement and integration of RHD control into primary care[185]. Integrated primary healthcare including task sharing with different cadres of healthcare providers will assist in delivering essential therapy and support disease control strategies. Advocacy efforts with governments, although challenging, must continue; starting with intensive disease control strategies for regions with high disease burden, using sociodemographic indices as surrogates until more accurate estimates are available.

In the next decade or so, sustained thrust towards finding ways to improve access to tertiary cardiovascular care for clinically manifest RHD and meaningful and sustained engagement of people with RHD is warranted. Efforts to build capacity within the most affected regions of the world and thoughtful integration of RHD into universal health coverage programmes are required. Gaps in preclinical, epidemiological, clinical and implementation science need to be addressed with comprehensive research, partnerships with lived experience experts and an all-of-society approach (Boxes 4 and 5).

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